

Summary of the “Decreasing Universe Theory”

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Introduction

The Decreasing Universe theory (D.U.T.) proposes that the gravitational field continuously contracts space and everything within it — rulers, atoms, and clocks alike.

This single effect provides an alternative explanation for two of the central puzzles of modern cosmology:

- The apparent accelerated expansion of the universe, without requiring dark energy.
- The anomalous orbital velocities of stars in galaxies, without invoking dark matter.

At first glance this idea may seem counterintuitive, but relativity already offers two analogous effects:

- Equivalence Principle — in an accelerating elevator, space and the objects inside it undergo continuous contraction in the direction of motion.
- Schwarzschild Metric — near a massive body, radial distances are altered, so that space becomes “compressed.”

Although the D.U. theory is not directly derived from these two effects, they illustrate that gravitationally induced contraction is not an unreasonable concept.

Dark Energy and the Expansion of the Universe

The D.U. theory explains the apparent accelerated expansion of galaxies as a consequence of local space contraction. Since we are ourselves within a gravitational field, the instruments we use to measure the universe — including atomic clocks and rulers — are also shrinking.

As a result, when we observe distant galaxies, they appear to be receding faster than expected, even if their actual motion remains unchanged. This contraction-based interpretation allows Hubble's Law to be derived directly, offering an alternative to the dark-energy hypothesis, and it further yields a closed-form expression for the distance of a galaxy as a function of its redshift [[01](#), [02](#)].

Dark Matter and Galactic Rotation Curves

The “Dark Matter” effect is traditionally invoked to explain why stars in the outer regions of galaxies move faster than expected from the visible mass alone. The D.U. theory offers an alternative explanation: continuous galactic contraction due to gravity.

- Closer to the galactic center, the gravitational field — and therefore the contraction rate — is higher, producing a greater reduction in emitted photon wavelengths.
- Farther from the center, the contraction rate decreases, so photon wavelengths are less compressed.

Since stellar velocities are measured through redshift (z), this differential contraction mimics an increased Doppler shift for stars farther from the galactic center — making their velocities appear higher than expected, without requiring an invisible “dark matter” component.

This mechanism has since been formalized through the Jocaxian Factor, which quantifies the contraction rate as a function of the galaxy's mass and the distance of a star from its center:

$$F_j(\Delta t, M, R) = e^{(J_0 \cdot M \cdot \Delta t / R^2)}$$

where J_0 is the Jocaxian Constant ($J_0 = H_0 \cdot R_t^2 / M_t \approx 1.37 \times 10^{-18} \text{ m}^2 \cdot \text{s}^{-1} \cdot \text{kg}^{-1}$, calibrated on the Milky Way's mass and the Hubble constant). The wavelength of a star at distance R from the galactic center then follows $\lambda_R = \lambda_0 / F_j(\Delta t, M, R)$, reproducing the observed rotation curves without a dark-matter halo [05].

The Tully-Fisher Relation as a Derived Consequence

Starting from this same formula for the dark-matter illusion, the Tully-Fisher relation — the well-known empirical correlation between a spiral galaxy's luminosity and its rotational velocity — can be derived directly within the D.U. framework, rather than assumed as an independent empirical law [06].

Galaxies Without “Dark Matter”: A Direct Test

Because the D.U. model attributes the dark-matter effect to the gravitational gradient between the central and peripheral stars of a galaxy, it predicts that this effect should be very small in two limiting cases: (a) ultra-diffuse galaxies with no dense massive core, and (b) extremely compact galaxies where all stars remain close to the center despite a large total mass.

Both cases are observed in nature. The ultra-diffuse galaxies NGC 1052-DF2, DF4, DF9, and AGC 114905 lack a defined nucleus and are consistently reported with little or no dark matter. The compact relic galaxy NGC 1277, despite hosting one of the most massive black holes known, shows rotation fully explained by its visible mass. Both outcomes are consistent with the gradient-based mechanism proposed by the D.U. theory [08].

Estimating the Age of the Universe from the CMB

The same contraction framework has been applied to the Cosmic Microwave Background: using the average wavelength of the CMB radiation, the D.U. theory yields an independent estimate for the age of the universe, without appealing to the standard hot Big Bang expansion history [07].

Testing the Theory Against the Λ CDM Model

The current standard cosmological model, Λ CDM, postulates Dark Energy and Dark Matter to explain these observed phenomena. The Decreasing Universe theory, in contrast, makes a directly testable prediction:

- Among galaxies at the same distance from Earth, more massive galaxies should show a lower redshift, due to their stronger gravitational contraction.

- The Λ CDM model predicts the opposite, since it assumes a gravitational redshift effect that increases with galaxy mass.

A dataset comparing galaxy mass and redshift was compiled to test this prediction, and an independent AI-based analysis was later run on the same data. Both analyses support the Decreasing Universe model and contradict the Λ CDM prediction [03, 09].

Independent Evidence: Time Dilation in Supernova Light Curves

A further line of evidence relevant to the D.U. theory comes from cosmic time dilation observed in Type Ia supernova (SN Ia) light curves. Distant supernovae take systematically longer to rise and fade than nearby ones. In the largest sample analyzed to date — 1504 SNe Ia from the Dark Energy Survey, spanning $0.1 \lesssim z \lesssim 1.2$ — the light-curve width was measured to scale as $(1+z)^b$, with $b = 1.003 \pm 0.005$ (stat) ± 0.010 (sys), closely matching the $(1+z)$ stretching conventionally attributed to cosmic expansion [10].

In a 2025 paper published in the journal *Galaxies* (MDPI), Vavryčuk argues that the standard FLRW metric, under the conventional assumption that its lapse function is fixed to unity, does not self-consistently account for this time dilation, and proposes a modified “Conformal Cosmology” metric that reproduces both the redshift and the time-dilation signal without requiring dark energy [12]. This is an independent critique, external to the D.U. framework, but it is consonant with the D.U. thesis: that the redshift-distance relation observed in SNe Ia can be explained without invoking dark energy, and that the conventional treatment of cosmic time and expansion in Λ CDM merits closer scrutiny.

Conclusion

The Decreasing Universe model offers a unified explanation for a growing set of independent observations: the apparent acceleration of cosmic expansion (without dark energy), galactic rotation curves (without dark matter), the Tully-Fisher relation, the existence of galaxies with little or no dark matter, and an independent estimate of the universe's age from the CMB.

By reconsidering the role of gravitational contraction as a single underlying mechanism, the theory offers a parsimonious alternative to Λ CDM — one that challenges the need for undetected exotic components such as dark matter and dark energy. The complete set of articles is maintained and updated at the D.U. archive [12].

References

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